

Chapter 5.2: Memory Technologies

SRAM, DRAM, Flash, and Disk

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1 Introduction: Four Memory Technologies

There are **four primary technologies** used in modern memory hierarchies:

1. **SRAM** (Static Random Access Memory) – Used for caches (closest to processor)
2. **DRAM** (Dynamic Random Access Memory) – Used for main memory
3. **Flash Memory** – Used for secondary storage in mobile devices
4. **Magnetic Disk** – Used for the largest/slowest level in servers

1.1 Comparison Table (2012 Data)

Memory Technology	Typical Access Time	Cost per GiB
SRAM	0.5–2.5 ns	\$500–\$1000
DRAM	50–70 ns	\$10–\$20
Flash	5,000–50,000 ns	\$0.75–\$1.00
Magnetic Disk	5,000,000–20,000,000 ns	\$0.05–\$0.10

Table 1: Access time and cost comparison of memory technologies (2012).

Key Trade-off

- **Faster** = More expensive per bit, smaller capacity
- **Cheaper** = Slower access, larger capacity

Notice how DRAM is $\sim 100\times$ slower than SRAM, but $\sim 50\times$ cheaper per bit!

2 SRAM Technology

SRAM – Static Random Access Memory

Used for: Caches (L1, L2, L3) integrated on the processor chip.

Key Characteristics:

- **Fixed access time** to any datum (random access)
- **No refresh needed** – value is kept as long as power is applied
- Access time $\approx$ cycle time (very efficient)
- Uses **6–8 transistors per bit** to prevent data disturbance during reads
- Only needs **minimal power** in standby mode

2.1 Why 6–8 Transistors?

The multiple transistors form a **bistable latch** that actively holds the stored bit. This design:

- Prevents the information from being disturbed when read
- Eliminates the need for refresh cycles
- Provides very fast access (< 3 ns)

2.2 Modern SRAM Usage

Thanks to **Moore's Law**, all levels of caches are now **integrated onto the processor chip**. The market for independent SRAM chips has nearly evaporated.

3 DRAM Technology

DRAM – Dynamic Random Access Memory

Used for: Main memory.

Key Characteristics:

- Value stored as **charge in a capacitor**
- Uses only **1 transistor per bit** (much denser than SRAM)
- **Must be refreshed** periodically (charge leaks away)
- Much cheaper per bit than SRAM

3.1 Why “Dynamic”?

The stored charge in a capacitor **cannot be kept indefinitely** – it leaks away over time (several milliseconds). The memory must be **periodically refreshed**:

Refresh Process

To refresh a DRAM cell:

1. **Read** the contents of the cell
2. **Write** the same value back

This restores the charge in the capacitor.

3.2 Efficient Refresh: Two-Level Decoding

DRAMs use a **two-level decoding structure** that allows refreshing an **entire row** at once:

- All cells in a row share a **word line**
- One read cycle + one write cycle refreshes the entire row
- Much more efficient than refreshing each bit individually

3.3 DRAM Internal Organization

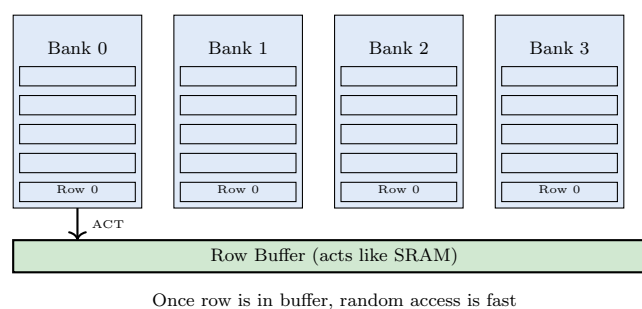


Figure 1: **Figure 5.4:** Internal organization of a DRAM with multiple banks.

Commands:

- **PRE (Precharge):** Opens or closes a bank
- **ACT (Activate):** Transfers a row to the row buffer
- Once in the buffer, data can be accessed at the width of the DRAM (4, 8, or 16 bits)

3.4 Synchronous DRAM (SDRAM)

SDRAM Advantage

Adding a **clock** to DRAM creates **Synchronous DRAM (SDRAM)**:

- Eliminates time for memory and processor to synchronize
- Clock transfers successive bits in a **burst** without additional address bits

3.5 Double Data Rate (DDR)

DDR SDRAM

Double Data Rate (DDR) means data is transferred on **both the rising and falling edges** of the clock.

Result: Twice the bandwidth for the same clock rate!

Example: DDR4-3200

- 3200 million transfers per second
- Clock frequency = $3200/2 = 1600$ MHz

3.6 Address Interleaving

To sustain high bandwidth, DRAMs use **multiple banks**:

Address Interleaving

Sending an address to several banks permits them all to read/write simultaneously.
With 4 banks:

- One initial access time
- Then accesses **rotate** between the four banks
- Result: $4\times$ **the bandwidth**

3.7 DIMMs and Memory Ranks

DIMM (Dual Inline Memory Module)

- Small board containing **4–16 DRAM chips**
- Typically organized to be **8 bytes wide** for servers
- Named by bandwidth: e.g., **PC25600** = 25,600 MB/s

Calculation: $\text{DDR4-3200} \times 8 \text{ bytes} = 25,600 \text{ MB/s}$

Memory Rank

A **memory rank** is the subset of chips in a DIMM that share common address lines and are used for a particular transfer.

4 Flash Memory

Flash Memory

Used for: Secondary storage in Personal Mobile Devices (PMDs), SSDs.

Key Characteristics:

- Type of **EEPROM** (Electrically Erasable Programmable Read-Only Memory)
- **Nonvolatile** – retains data without power
- **Writes can wear out** the memory bits over time
- Much faster than magnetic disks ($\sim 1000\times$)

4.1 Write Wear and Wear Leveling

Write Wear-Out Problem

Unlike DRAM and disks, flash memory bits can be **worn out** by repeated writes. Each cell has a limited number of write cycles (typically 10,000–100,000).

Wear Leveling

To cope with write limits, flash controllers use **wear leveling**:

- Spreads writes across all blocks
- Remaps frequently written blocks to less-used blocks
- Ensures even wear distribution
- Can also map out defective cells to improve yield

4.2 Flash Advantages

- **No mechanical parts** – very rugged
- Perfect for mobile devices that experience jostling
- Lower power consumption than disks
- Fast random access compared to disks

5 Magnetic Disk Technology

Magnetic Disk (Hard Disk Drive)

Used for: Largest/slowest level in server memory hierarchies.

Key Characteristics:

- **Mechanical device** with rotating platters
- Very high capacity at low cost
- Very slow access time (milliseconds)
- **No write wear-out** problem

5.1 Physical Structure

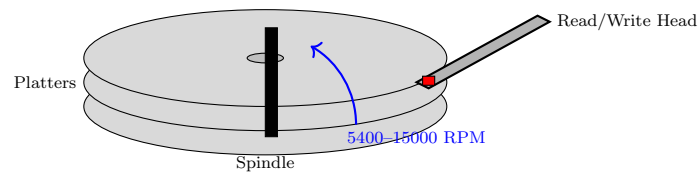


Figure 2: **Figure 5.6:** A magnetic hard disk with platters and read/write head.

Components:

- **Platters:** Metal disks covered with magnetic recording material
- **Spindle:** Rotates platters at 5400–15,000 RPM
- **Read-Write Head:** Electromagnetic coil on movable arm
- Drive is **sealed** to control environment and allow closer head positioning

5.2 Disk Organization: Tracks, Sectors, Cylinders

Track

One of **thousands of concentric circles** that make up the surface of a magnetic disk.

Sector

One of the **segments that make up a track**. A sector is the **smallest unit** of information that can be read or written (typically 512–4096 bytes).

Cylinder

The set of all tracks under the read-write heads at a given point **on all surfaces**.

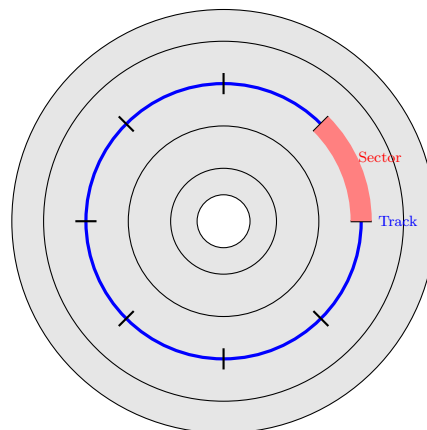


Figure 3: Top view of a disk surface showing tracks and sectors.

5.3 Three Stages of Disk Access

1. **Seek Time:** Position head over the correct track
2. **Rotational Latency:** Wait for desired sector to rotate under head
3. **Transfer Time:** Read/write the actual data

Seek Time

The time to move the read/write head to the desired track.

- Advertised average: 3–13 ms
- **Actual average** may be 25–33% of advertised due to **locality of disk references**

Rotational Latency

The time for the desired sector to rotate under the head.

- **Average latency** = Half a full rotation
- At 5400 RPM: $\frac{1}{5400} \times 60 \times 0.5 = 5.56$ ms
- At 15000 RPM: $\frac{1}{15000} \times 60 \times 0.5 = 2$ ms

Transfer Time

Time to transfer the block of bits. Function of:

- Sector size
- Rotation speed
- Recording density

Transfer rates: 100–200 MB/sec (2012)

5.4 Disk vs. Semiconductor Memory

	Disk	Flash	DRAM
Access Time	Slowest (ms)	1000× faster	100,000× faster
Cost/Bit	Cheapest	10–100× more	100–1000× more
Nonvolatile	Yes	Yes	No
Write Wear	No	Yes	No
Ruggedness	Poor	Excellent	N/A

Table 2: Comparison of disk, flash, and DRAM technologies.

6 Flashcards

1. **Q: What are the four primary technologies used in modern memory hierarchies?**
A: DRAM (Dynamic Random Access Memory), SRAM (Static Random Access Memory), Flash memory, and Magnetic Disk.
2. **Q: What type of memory is typically used to implement main memory?**
A: DRAM (Dynamic Random Access Memory).
3. **Q: What type of memory is typically used for caches closest to the processor?**
A: SRAM (Static Random Access Memory).
4. **Q: Which is less costly per bit, DRAM or SRAM?**
A: DRAM is less costly per bit than SRAM.
5. **Q: What is the primary reason for the price difference between DRAM and SRAM?**
A: DRAM uses significantly less area per bit of memory, allowing for larger capacity on the same amount of silicon.
6. **Q: What nonvolatile memory technology is commonly used as secondary memory in Personal Mobile Devices?**
A: Flash memory.
7. **Q: In servers, what technology is used to implement the largest and slowest level in the memory hierarchy?**
A: Magnetic disk.
8. **Q: What is the typical access time for SRAM?**
A: 0.5–2.5 nanoseconds.
9. **Q: What is the typical access time for DRAM?**
A: 50–70 nanoseconds.
10. **Q: What is the typical access time for Flash memory?**
A: 5,000–50,000 nanoseconds.
11. **Q: What is the typical access time for a magnetic disk?**
A: 5,000,000–20,000,000 nanoseconds.
12. **Q: Why is the access time for SRAM very close to its cycle time?**
A: Because SRAMs do not need to be refreshed.
13. **Q: How many transistors per bit do SRAMs typically use?**
A: Six to eight transistors per bit.
14. **Q: What is the primary function of the 6–8 transistors in an SRAM cell?**
A: To prevent the stored information from being disturbed when read.
15. **Q: Due to Moore's Law, where are all levels of caches typically integrated today?**
A: They are integrated onto the processor chip.
16. **Q: How is a value stored in a DRAM cell?**
A: As a charge in a capacitor.
17. **Q: How many transistors are used per bit of storage in a DRAM cell?**
A: A single transistor.
18. **Q: Why is DRAM so much denser and cheaper per bit than SRAM?**
A: Because DRAMs use only one transistor per bit of storage, compared to SRAM's six to eight.
19. **Q: Why is DRAM called 'dynamic'?**
A: Because the charge stored in its capacitor cells cannot be kept indefinitely and must be periodically refreshed.
20. **Q: What is the process for refreshing a DRAM cell?**
A: Its contents are read and then written back.

21. **Q: How does the two-level decoding structure in DRAMs facilitate efficient refreshing?**
A: It allows an entire row to be refreshed at once with a single read cycle followed by a write cycle.
22. **Q: In a modern DRAM's internal organization, what command is sent to open or close a bank?**
A: A PRE (precharge) command.
23. **Q: In a modern DRAM, what command causes a row to be transferred to a buffer?**
A: An Act (activate) command, sent with the row address.
24. **Q: How does a DRAM improve performance for repeated accesses to the same row?**
A: It buffers the entire row, and this buffer acts like an SRAM for faster access.
25. **Q: What is the primary advantage of Synchronous DRAMs (SDRAMs)?**
A: The use of a clock eliminates the time needed for the memory and processor to synchronize.
26. **Q: How do SDRAMs achieve a speed advantage for burst transfers?**
A: The clock transfers successive bits in a burst without needing additional address bits.
27. **Q: What does DDR stand for in DDR SDRAM?**
A: Double Data Rate.
28. **Q: What is the key mechanism behind Double Data Rate (DDR) technology?**
A: Data is transferred on both the rising and falling edges of the clock, effectively doubling the bandwidth.
29. **Q: A DDR4-3200 DRAM can perform 3200 million transfers per second. What is its clock frequency?**
A: 1600 MHz.
30. **Q: Define Address Interleaving.**
A: A rotating access scheme where an address is sent to several banks, allowing them all to read or write simultaneously to supply higher bandwidth.
31. **Q: What is a DIMM?**
A: A Dual Inline Memory Module – a small board containing multiple DRAM chips, commonly used for server memory.
32. **Q: A DIMM using DDR4-3200 SDRAMs, organized to be 8 bytes wide, would be named what?**
A: PC25600 (based on its bandwidth of 25,600 MB/s).
33. **Q: Define Memory Rank.**
A: A subset of chips in a DIMM that share common address lines.
34. **Q: Flash memory is a type of ____.**
A: Electrically erasable programmable read-only memory (EEPROM).
35. **Q: What is a primary limitation of flash memory regarding write operations?**
A: Writes can wear out the flash memory bits over time.
36. **Q: Define Wear Leveling.**
A: A technique used by flash memory controllers to spread writes evenly by remapping frequently written blocks to less used blocks.
37. **Q: What is the main purpose of wear leveling in flash memory products?**
A: To prevent premature failure of the memory by distributing the wear from write operations across all blocks.
38. **Q: What are the rotating components of a magnetic hard disk called?**
A: Platters.
39. **Q: What is the component in a hard disk that contains a small electromagnetic coil to read and write information?**
A: A read-write head.
40. **Q: Define Track (on a magnetic disk).**
A: One of thousands of concentric circles that make up the surface of a magnetic disk.

41. **Q: Define Sector (on a magnetic disk).**
A: One of the segments that make up a track; it is the smallest amount of information that is read or written on a disk.
42. **Q: Define Cylinder (on a magnetic disk).**
A: The set of all tracks under the read-write heads at a given point on all surfaces.
43. **Q: What is the first stage in a disk access process, involving positioning the head over the proper track?**
A: A seek.
44. **Q: Define Seek Time.**
A: The time required to move the read-write head to the desired track on a disk.
45. **Q: Why might the actual average seek time be much lower than the advertised average seek time?**
A: Due to the locality of disk references, where successive accesses are often to the same file or nearby tracks.
46. **Q: Define Rotational Latency.**
A: The time required for the desired sector of a disk to rotate under the read-write head.
47. **Q: What is the average rotational latency generally assumed to be?**
A: Half the time of a full rotation.
48. **Q: What are the three main stages of a disk access process?**
A: Seek time, rotational latency, and transfer time.
49. **Q: What is the final component of a disk access time?**
A: Transfer time – the time to transfer the actual block of bits.
50. **Q: Transfer time is a function of what three factors?**
A: Sector size, rotation speed, and the recording density of a track.
51. **Q: Compared to magnetic disks, approximately how much faster is DRAM?**
A: DRAM is approximately 100,000 times faster than a magnetic disk.
52. **Q: What is a key advantage of magnetic disks over flash memory regarding data writes?**
A: Magnetic disks do not have a write wear-out problem.
53. **Q: Why is flash memory a better match than magnetic disks for personal mobile devices?**
A: Flash memory is much more rugged and can better withstand the jostling inherent in mobile devices.